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# Learning and Assignment of IP Address manually to computer

**Aim:** To Learn IP addressing & Manually assign are IP address to a computer To verify the assigned IP address using Linux networking commands

**Theory:**

An IP Address is a unique numerical address assigned to every device connected to a network. It enables communication blw devices over a network. An IPv4 address consist of four octlets seperated by dots (eg: 192.168.1.10)

An IP Address can be Assigned either Manually (static IP) or automatically (Dynamic IP) using DHCP. A manually assigned IP address remains unless changed by the User along with the IP address remains when parameters such as the Subnet Mask, default gateway & DNS server are also configured for proper network.

## Commands & Sample Outputs:

1. Display all IP address

command:

ip address show

Sample-output:

2: ens160:
inet 172.16.211.128/24
inet6 Fc80::20c:29FF:Fe75:9927/64

Observation:

Displays the $IPv_4$ and $IPv_6$ address of all networks.

2. Display a specific network interface

command

ip addr show dev ens160

Sample-Output:

2: ens160
inet 172.16.211.128/24

3. Display Routing table

command

ip route show

Sample Output

default via 172.16.211.2 dev ens160
172.16.211.0/24 dev ens160

observation:- Displays the routing Table and the default gateway

4. Display Neighbour (ARP) Table :-

command

ip neigh show

sample output

172.16.211.1 REACHABLE
172.16.211.254 STABLE

observation:- Displays the neighbouring devices and their MAC address

Result:- The IP address was manually assigned and verified successfully using Linux networking commands.